2. Build an Artificial Neural Network by implementing the Forward propagation algorithm and test the same using appropriate data sets.

```
import numpy as np

# Step 1: Define activation function (Sigmoid)
def sigmoid(x):
    return 1 / (1 + np.exp(-x))

# Step 2: Input dataset (Example: 2 features)
# Each row = one data sample
X = np.array([
    [0, 0],
    [0, 1],
    [1, 0],
    [1, 1]
])

# Step 3: Initialize weights and biases (random values)
np.random.seed(1)

# Input layer (2 neurons) -> Hidden layer (2 neurons)
W1 = np.random.rand(2, 2)
b1 = np.random.rand(1, 2)

# Hidden layer (2 neurons) -> Output layer (1 neuron)
W2 = np.random.rand(2, 1)
b2 = np.random.rand(1, 1)

# Step 4: Forward Propagation
def forward_propagation(X):
    # Input to Hidden Layer
    Z1 = np.dot(X, W1) + b1
    A1 = sigmoid(Z1)

    # Hidden to Output Layer
    Z2 = np.dot(A1, W2) + b2
    A2 = sigmoid(Z2)

    return A2

# Step 5: Test the model
output = forward_propagation(X)

print("Input:")
print(X)

print("\nOutput after Forward Propagation:")
print(output)
```